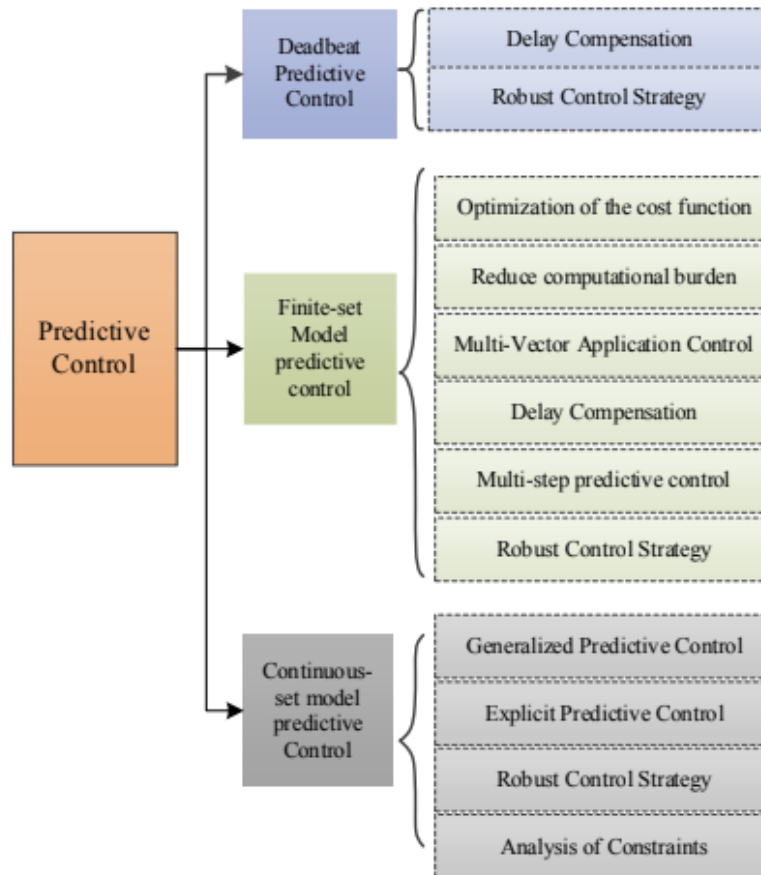


Review Paper Summary

There were 2 traditional methods of control: vector control and direct torque control. The PI controller in vector control includes multiple intermediate variables in the parameter tuning process and the direct torque control has a large torque ripple and introduces high frequency noise.

The Predictive Control Technology can be divided into 3 categories-

- a. Deadbeat Predictive Control
- b. Continuous-Set Model Predictive Control
- c. Finite-Set Model Predictive Control



Algorithm	Pros	Cons
Deadbeat Predictive Control	1. Keeps the system harmonic free 2. Fixed switching frequency. 3. High precision and lower calculation amount.	1. Slight difference in parameters affects the control performance. 2. Real time control requires delay compensations.
Finite-Set Model	1. Can control multivariable and	1. Variable switching frequency,

Predictive Control	nonlinear systems 2. Can predict overvoltage and overcurrent	harmonics will be introduced to the system. 2. As the number of possible voltage vectors increases, the delay also increases. 3. Implementation is complex, but doesn't require modulation state
Continuous-Set Model Predictive Control	1. Can control multivariable and nonlinear systems 2. Can predict overvoltage and overcurrent 3. Fixed switching frequency	1. Requires a modulation stage, adding complexity. 2. Requires higher computational efforts. 3. Implementation complexity increases with multiple control objectives

All predictive control algorithms rely on an accurate mathematical model of the motor. Any change in motor parameters (e.g., resistance, inductance, flux linkage) can lead to incorrect predictions, affecting control performance and robustness. Model mismatches may cause errors in current, torque, and speed regulation. Robust control strategies or online parameter estimation can help mitigate this issue.

Future research possibilities

1. Optimization of the weights of each component in the cost function using ANN.
2. Modifying the cost function for reduction of the switching losses.
3. Combining both, including the switching loss factor with optimization of the weights using ANN.
4. Control performance comparison of PI, MPC (FCS-MPC), ANN and RL(Reinforcement Learning) models. (Completed for PI and MPC)
5. Using Neural Networks for online parameter estimation to improve robustness of the system

My viewpoints: I suggest considering either the third, fourth or fifth option. For the third option, if we can identify a cost function component to minimize switching losses, integrating it with an ANN model for weight optimization would primarily involve data processing and model training. As for the fourth option, since we previously selected this topic and have already completed the MPC comparison, transitioning to ANN model training would be a natural next step. For the fifth option, we just need to train the ANN using a dataset on parameter estimation ensuring that the model doesn't overfit or underfit.